

Image and video processing.

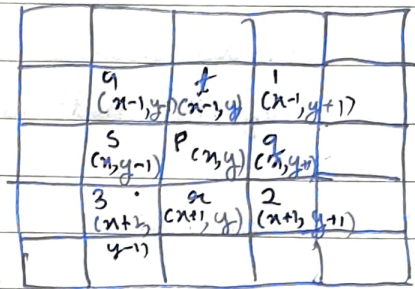
Some of the basic relationship between pixels.

(i) Neighbours of pixel.

→ Neighbours of pixel

$$N_4(p) = \{q, r, s, t\}$$

$$= \{(x, y+1), (x+1, y), (x, y-1), (x-1, y)\}$$



$$N_D(P) = \{1, 2, 3, 4\}$$

$$= \{(x-1, y+1), (x+1, y+1), (x+1, y-1), (x-1, y-1)\}$$

$$N_8(P) = N_4(P) \cup N_D(P)$$

$$= \{(x, y+1), (x+1, y), (x, y-1), (x-1, y), (x-1, y+1), (x+1, y+1), (x+1, y-1), (x-1, y-1)\}$$

→ Adjacency, connectivity, region & boundary

(a) 4-adjacency, b) 8-adjacency, c) m-adjacency

Condition for adjacency:-

(i) Suppose the image has 2^n level ($L = 2^n$), then adjacent pixels should have same value in subset $V = \{1, \dots, n\}$

ii) 4-adjacency: A pixel is 4-adjacent to P if it satisfies conditions (i) and q is a $m_4(x)$ 4-neighbour of P

(iii) 8-adjacency conditions:

A pixel q is 8-adjacent to p if it satisfies condition (i) and q must be in $N_8(p)$

(iv) m-adjacency condition: intersection of $N_4(p)$ and $N_8(p)$ not containing the value in subset V

→ Connectivity

* horizontal, vertical path

- 4 adj path

* diagonal :- 8 adjacent path

→ 4-adjacent path is preferred over 8-adjacent path

m-adj path

0	1	1
0	1	0
0	0	1

→ Regions

Two regions are connected if there exists a path between two regions, otherwise they are known as disjoint regions.

0	1	1	} R_i
0	1	0	
1	0	0	
0	1	0	} R_j
1	1	1	
1	1	1	

→ Boundary: Boundary is the regions where region have one boundary pixel or boundary pixel.

→ Distance between pixel:

let 3 pixels be $P(x, y)$, $Q(s, t)$, $R(m, n)$.

a) $d(P, Q) \geq 0$, $d(P, Q) = 0$
 $\therefore P = Q$

(b) $d(P, Q) = d(Q, P)$

(c) $d(P, R) \leq d(P, Q) + d(Q, R)$
 equal to condition occur ~~men~~ when
 $d(P, Q) / d(Q, R) = 0$

d) Euclidean distance

$$d(P, Q) = \sqrt{(x-s)^2 + (y-t)^2}$$

e) 4-distance (city block distance)

$$d_4(P, Q) = \{ |x-s| + |y-t| \}$$

$$d_4(P, Q) = 1$$

$$d_4(P, R) = 2$$

$$\begin{matrix} & 2 & 1 & 2 \\ 2 & 1 & 0 & 1 & 2 \\ & 2 & 1 & 2 \end{matrix}$$

(f) 8-distance : (chess board distance)

$$d_8(P, Q) = \max \{ |x-s| + |y-t| \}$$

$$\begin{matrix} 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 1 & 1 & 2 \\ 2 & 1 & 0 & 1 & 2 \\ 2 & 1 & 1 & 1 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{matrix}$$

→ Linear versus non-linear operations

$$f(x, y) \rightarrow \boxed{H} \rightarrow g(x, y)$$

(operation)

$$(i) H\{f(x, y)\} = g(x, y) \quad \text{--- (i)}$$

$$\begin{aligned} (ii) H\{a f_1(x, y) + b f_2(x, y)\} \\ &= a H\{f_1(x, y)\} + b H\{f_2(x, y)\} \\ &= a g_1(x, y) + b g_2(x, y) \end{aligned}$$

If LHS = RHS, then H is a linear operation ~~ex.~~

(iii) Non-linear operation

$H \rightarrow$ maximum operator

$$f_1 = \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix}, f_2 = \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix} \quad a=1, b=-1$$

$$\max [a f_1(x, y) + b f_2(x, y)] = \cancel{1(3)} - \cancel{1(7)} = -4 \neq \text{RHS}$$

$$\bullet \max \left[1 \cdot \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} \right] - 1 \cdot \max \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix} = 1(3) - 1(7) = -4 = \text{RHS}$$

LHS =

$$\begin{aligned} &\max \left[1 \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} - 1 \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix} \right] \\ &= \max \begin{bmatrix} -6 & -3 \\ -2 & -4 \end{bmatrix} = -2 \neq -4 \end{aligned}$$

$\therefore \text{LHS} \neq \text{RHS}$

So, max is a non-linear operator.